

Gra Snake -- Technika Mikroprocesorowa

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Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

fruit	Owoc dla weza, zjedzenie zwieksza dl weza	??
snake	Snake tj tablica segmentow + aktualna dlugosc + kierunek	??
snakeSeg	Klasa segment weza	??

Chapter 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

fruit.cpp		??
fruit.h		??
snake.cpp		??
snake.h		??

Chapter 3

Class Documentation

3.1 fruit Struct Reference

owoc dla weza, zjedzenie zwieksza dl weza

```
#include <fruit.h>
```

Public Attributes

- int [siz](#) =5
- int [heig](#) =30
- int [widt](#) =30
- int [padd](#) =10

3.1.1 Detailed Description

owoc dla weza, zjedzenie zwieksza dl weza

Definition at line [10](#) of file [fruit.h](#).

3.1.2 Member Data Documentation

3.1.2.1 heig

```
int fruit::heig =30
```

Definition at line [13](#) of file [fruit.h](#).

3.1.2.2 padd

```
int fruit::padd =10
```

Definition at line [15](#) of file [fruit.h](#).

3.1.2.3 siz

```
int fruit::siz =5
```

Definition at line 12 of file [fruit.h](#).

3.1.2.4 widt

```
int fruit::widt =30
```

Definition at line 14 of file [fruit.h](#).

The documentation for this struct was generated from the following file:

- [fruit.h](#)

3.2 snake Struct Reference

snake tj tablica segmentow + aktualna dlugosc + kierunek

```
#include <snake.h>
```

Public Attributes

- [snakeSeg segs](#) [[SnMAXsiz](#)]
- int [CurrLen](#) =3
- int [dir](#) = 0

3.2.1 Detailed Description

snake tj tablica segmentow + aktualna dlugosc + kierunek

Definition at line 22 of file [snake.h](#).

3.2.2 Member Data Documentation

3.2.2.1 CurrLen

```
int snake::CurrLen =3
```

Definition at line 24 of file [snake.h](#).

3.2.2.2 dir

```
int snake::dir = 0
```

Definition at line 25 of file [snake.h](#).

3.2.2.3 segs

```
snakeSeg snake::segs[SnMAXsiz]
```

Definition at line 23 of file [snake.h](#).

The documentation for this struct was generated from the following file:

- [snake.h](#)

3.3 snakeSeg Struct Reference

klasa segment weza

```
#include <snake.h>
```

Public Attributes

- int [heig](#) =0
- int [widt](#) =0

3.3.1 Detailed Description

klasa segment weza

Definition at line 13 of file [snake.h](#).

3.3.2 Member Data Documentation

3.3.2.1 heig

```
int snakeSeg::heig =0
```

Definition at line 15 of file [snake.h](#).

3.3.2.2 widt

```
int snakeSeg::widt =0
```

Definition at line 16 of file [snake.h](#).

The documentation for this struct was generated from the following file:

- [snake.h](#)

Chapter 4

File Documentation

4.1 fruit.cpp File Reference

```
#include "fruit.h"
```

Functions

- void `setFruit` (`fruit` &Obj)
wylosuj nowa pozycje weza
- bool `ateFruit` (`snake` &oSnake, `fruit` &oFruit)
zwieksz rozmiar weza jezeli zje owoc

4.1.1 Function Documentation

4.1.1.1 ateFruit()

```
bool ateFruit (  
    snake & oSnake,  
    fruit & oFruit)
```

zwieksz rozmiar weza jezeli zje owoc

Definition at line 13 of file `fruit.cpp`.

```
00013                                     {  
00014     if((oSnake.segs[0].widt <= oFruit.widt + oFruit.siz) && (oSnake.segs[0].widt >= oFruit.widt -  
00015         oFruit.siz) ) {  
00016         if ((oSnake.segs[0].heig <= oFruit.heig + oFruit.siz) && (oSnake.segs[0].heig >= oFruit.heig -  
00017             oFruit.siz)) {  
00018             //jak zwiekszac rozmiar weza??  
00019             oSnake.CurrLen++;  
00020             if(oSnake.CurrLen >= SnMAXsiz) {  
00021                 return true;  
00022             }  
00023             setFruit(oFruit);  
00024         }  
00025     }  
00026     return false;  
00027 }
```

4.1.1.2 setFruit()

```
void setFruit (
    fruit & Obj)
```

wylosuj nowa pozycje weza

Definition at line 6 of file [fruit.cpp](#).

```
00006 {
00007     Obj.heig=rand()%(SCREEN_HEIGHT-2*Obj.padd)+Obj.padd;
00008     // =random(SCREEN_HEIGHT-padd)+(padd/2);
00009     Obj.widt=rand()%(SCREEN_WIDTH-2*Obj.padd)+Obj.padd;
00010 }
```

4.2 fruit.cpp

[Go to the documentation of this file.](#)

```
00001 #include "fruit.h"
00002
00003
00004
00005
00006 void setFruit (fruit &Obj) {
00007     Obj.heig=rand()%(SCREEN_HEIGHT-2*Obj.padd)+Obj.padd;
00008     // =random(SCREEN_HEIGHT-padd)+(padd/2);
00009     Obj.widt=rand()%(SCREEN_WIDTH-2*Obj.padd)+Obj.padd;
00010 }
00011
00012 //bool?
00013 bool ateFruit(snake &oSnake, fruit &oFruit) {
00014     if((oSnake.segs[0].widt <= oFruit.widt + oFruit.siz) && (oSnake.segs[0].widt >= oFruit.widt -
00015         oFruit.siz) ) {
00016         if ((oSnake.segs[0].heig <= oFruit.heig + oFruit.siz) && (oSnake.segs[0].heig >= oFruit.heig -
00017             oFruit.siz)) {
00018             //jak zwiekszac rozmiar weza??
00019             oSnake.CurrLen++;
00020             if(oSnake.CurrLen >= SnMAXsiz) {
00021                 return true;
00022             }
00023             setFruit(oFruit);
00024         }
00025     }
00026     return false;
00027 }
```

4.3 fruit.h File Reference

```
#include "snake.h"
#include <stdlib.h>
```

Classes

- struct [fruit](#)
owoc dla weza, zjedzenie zwieksza dl weza

Functions

- void [setFruit](#) ([fruit](#) &Obj)
wylosuj nowa pozycje weza
- bool [ateFruit](#) ([snake](#) &oSnake, [fruit](#) &oFruit)
zwieksz rozmiar weza jezeli zje owoc

4.3.1 Function Documentation

4.3.1.1 ateFruit()

```
bool ateFruit (
    snake & oSnake,
    fruit & oFruit)
```

zwiększ rozmiar weza jezeli zje owoc

Definition at line 13 of file [fruit.cpp](#).

```
00013                                     {
00014     if((oSnake.segs[0].width <= oFruit.width + oFruit.siz) && (oSnake.segs[0].width >= oFruit.width -
oFruit.siz) ) {
00015         if ((oSnake.segs[0].heig <= oFruit.heig + oFruit.siz) && (oSnake.segs[0].heig >= oFruit.heig -
oFruit.siz)) {
00016             //jak zwiększac rozmiar weza??
00017             oSnake.CurrLen++;
00018             if(oSnake.CurrLen >= SnMAXsiz) {
00019                 return true;
00020             }
00021             setFruit(oFruit);
00022         }
00023     }
00024     return false;
00025 }
```

4.3.1.2 setFruit()

```
void setFruit (
    fruit & Obj)
```

wylosuj nowa pozycje weza

Definition at line 6 of file [fruit.cpp](#).

```
00006                                     {
00007     Obj.heig=rand()%(SCREEN_HEIGHT-2*Obj.padd)+Obj.padd;
00008     // =random(SCREEN_HEIGHT-padd)+(padd/2);
00009     Obj.width=rand()%(SCREEN_WIDTH-2*Obj.padd)+Obj.padd;
00010 }
```

4.4 fruit.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "snake.h"
00004
00005 #include <stdlib.h>
00006
00010 struct fruit{
00011     // uint_8
00012     int siz =5;
00013     int heig=30;
00014     int width=30; //pozycja segm
00015     int padd=10; //padding
00016 };
00017
00021 void setFruit (fruit &Obj);
00022
00026 //bool?
00027 bool ateFruit(snake &oSnake, fruit &oFruit);
00028
```

4.5 snake.cpp File Reference

```
#include "snake.h"
```

Functions

- void [setSeg](#) ([snake](#) &Obj, int seg, int aheig, int awidt)
ustaw parametry weza
- void [changeDir](#) ([snake](#) &Obj, bool ifRight)
zmien kierunek weza
- bool [goSnake](#) ([snake](#) &Obj)
1 krok weza w dobra strone
- void [addSegm](#) ([snake](#) &Obj, int aHei, int aWid)
dodaj segment weza
- void [moveSnake](#) ([snake](#) &Obj)
przesun pola tablicy segmentow o 1
- bool [selfColl](#) ([snake](#) &Obj)
kolizja ze soba

4.5.1 Function Documentation

4.5.1.1 addSegm()

```
void addSegm (  
    snake & Obj,  
    int aHei,  
    int aWid)
```

dodaj segment weza

Definition at line 62 of file [snake.cpp](#).

```
00062                                     {  
00063     if (Obj.CurrLen<SnMAXsiz) {  
00064         for (int i = Obj.CurrLen; i > 0; i--) {  
00065             Obj.segs[i] = Obj.segs[i - 1];  
00066         }  
00067         Obj.segs[0].heig=aHei;  
00068         Obj.segs[0].widt=aWid;  
00069         Obj.CurrLen++;  
00070     }  
00071 }
```

4.5.1.2 changeDir()

```
void changeDir (
    snake & Obj,
    bool ifRight)
```

zmien kierunek weza

Parameters

<i>Obj</i>	ref do weza
<i>ifRight</i>	czy skret w prawo

Definition at line 11 of file [snake.cpp](#).

```
00011                                     { //true -->right
00012     if (ifRight) {
00013         //go right
00014         Obj.dir = (Obj.dir + 1) % 4; //+1
00015     } else {
00016         //go left
00017         Obj.dir = (Obj.dir + 3) % 4; //-1
00018     }
00019 }
```

4.5.1.3 goSnake()

```
bool goSnake (
    snake & Obj)
```

1 krok weza w dobra strone

petle musza ustawiac nowy head i kopiowac wszystkie inne elem

Parameters

<i>Obj</i>	ref do snake
------------	--------------

Returns

if gameOver –tj. czy wyszedl poza ekran

Definition at line 24 of file [snake.cpp](#).

```
00024                                     {
00025     moveSnake (Obj);
00026
00027     switch (Obj.dir*3) {
00028
00029         case 12: //gora
00030         case 0 : //gora
00031             Obj.segs[0].heig--;
00032             // heig--; //na OLED -- idzie w gore,
00033             break;
00034
00035         case 3 : //prawo
00036             Obj.segs[0].widt++;
00037             // widt++;
00038             break;
00039
00040         case 6 : //dol
00041             Obj.segs[0].heig++;
00042             // heig++;
00043             break;
```

```

00044
00045     case 9 : //prawo
00046         Obj.segs[0].widt--;
00047         // widt--;
00048         break;
00049     }
00050
00051     //czy snake wyszedl poza ekran
00052     if(Obj.segs[0].heig > SCREEN_HEIGHT || Obj.segs[0].heig < 0)
00053         return true;
00054     else if(Obj.segs[0].widt>SCREEN_WIDTH || Obj.segs[0].widt < 0)
00055         return true;
00056     else {
00057         return false;
00058     }
00059     // displSnake(2);
00060 }

```

4.5.1.4 moveSnake()

```

void moveSnake (
    snake & Obj)

```

przesun pola tablicy segmentow o 1

Definition at line 73 of file [snake.cpp](#).

```

00073     {
00074         if(Obj.CurrLen<SnMAXsiz) {
00075             for (int i = Obj.CurrLen-1; i > 0; i--) {
00076                 Obj.segs[i] = Obj.segs[i - 1];
00077             }
00078         }
00079     }

```

4.5.1.5 selfColl()

```

bool selfColl (
    snake & Obj)

```

kolizja ze soba

Returns

true dla kollisioni – game Over;

najpierw dla każdego elementu porównujemy wysokość, zamiast wys && szer, dzięki temu możemy odciąć połowę operacji od razu – szybsze przetwarzanie

Definition at line 81 of file [snake.cpp](#).

```

00081     {
00082         for(int i=1; i<Obj.CurrLen-1; i++) {
00083             if(Obj.segs[0].heig==Obj.segs[i].heig ) {
00084                 if(Obj.segs[0].widt==Obj.segs[i].widt) {
00085                     return true;
00086                 }
00087             }
00088         }
00089         return false;
00090     }

```

4.5.1.6 setSeg()

```
void setSeg (
    snake & Obj,
    int seg,
    int aheig,
    int awidt)
```

ustaw parametry weza

Parameters

<i>Obj</i>	ref do weza
<i>seg</i>	numer segmentu weza
<i>aheig</i>	wysokosc do ustawienia
<i>awidt</i>	szer do ustawienia

Definition at line 4 of file [snake.cpp](#).

```
00004
00005     if(seg<SnMAXsiz) {
00006         Obj.segs[seg].heig=aheig;
00007         Obj.segs[seg].widt=awidt;
00008     }
00009 }
```

4.6 snake.cpp

[Go to the documentation of this file.](#)

```
00001 #include "snake.h"
00002
00003
00004 void setSeg(snake &Obj, int seg, int aheig, int awidt) {
00005     if(seg<SnMAXsiz) {
00006         Obj.segs[seg].heig=aheig;
00007         Obj.segs[seg].widt=awidt;
00008     }
00009 }
00010
00011 void changeDir (snake &Obj, bool ifRight) { //true -->right
00012     if (ifRight) {
00013         //go right
00014         Obj.dir = (Obj.dir + 1) % 4; //+1
00015     } else {
00016         //go left
00017         Obj.dir = (Obj.dir + 3) % 4; //-1
00018     }
00019 }
00020
00024 bool goSnake (snake &Obj) {
00025     moveSnake(Obj);
00026
00027     switch (Obj.dir*3) {
00028
00029         case 12: //gora
00030         case 0 : //gora
00031             Obj.segs[0].heig--;
00032             // heig--; //na OLED -- idzie w gore,
00033             break;
00034
00035         case 3 : //prawo
00036             Obj.segs[0].widt++;
00037             // widt++;
00038             break;
00039
00040         case 6 : //dol
00041             Obj.segs[0].heig++;
00042             // heig++;
00043             break;
```

```

00044
00045     case 9 : //prawo
00046         Obj.segs[0].widt--;
00047         // widt--;
00048         break;
00049     }
00050
00051     //czy snake wyszedl poza ekran
00052     if(Obj.segs[0].heig > SCREEN_HEIGHT || Obj.segs[0].heig < 0)
00053         return true;
00054     else if(Obj.segs[0].widt>SCREEN_WIDTH || Obj.segs[0].widt < 0)
00055         return true;
00056     else {
00057         return false;
00058     }
00059     // displSnake(2);
00060 }
00061
00062 void addSegm(snake &Obj, int aHei, int aWid) {
00063     if(Obj.CurrLen<SnMAXsiz) {
00064         for (int i = Obj.CurrLen; i > 0; i--) {
00065             Obj.segs[i] = Obj.segs[i - 1];
00066         }
00067         Obj.segs[0].heig=aHei;
00068         Obj.segs[0].widt=aWid;
00069         Obj.CurrLen++;
00070     }
00071 }
00072
00073 void moveSnake(snake &Obj) {
00074     if(Obj.CurrLen<SnMAXsiz) {
00075         for (int i = Obj.CurrLen-1; i > 0; i--) {
00076             Obj.segs[i] = Obj.segs[i - 1];
00077         }
00078     }
00079 }
00080
00081 bool selfColl (snake &Obj){
00082     for(int i=1; i<Obj.CurrLen-1; i++) {
00083         if(Obj.segs[0].heig==Obj.segs[i].heig ) {
00084             if(Obj.segs[0].widt==Obj.segs[i].widt) {
00085                 return true;
00086             }
00087         }
00088     }
00089     return false;
00090 }

```

4.7 snake.h File Reference

Classes

- struct [snakeSeg](#)
klasa segment weza
- struct [snake](#)
snake tj tablica segmentow + aktualna dlugosc + kierunek

Macros

- #define [SnMAXsiz](#) 10
- #define [SCREEN_WIDTH](#) 128
- #define [SCREEN_HEIGHT](#) 64

Functions

- void [setSeg](#) ([snake](#) &Obj, int seg, int aHeig, int aWid)
ustaw parametry weza
- void [changeDir](#) ([snake](#) &Obj, bool ifRight)
zmien kierunek weza
- void [addSegm](#) ([snake](#) &Obj, int aHei, int aWid)
dodaj segment weza
- void [moveSnake](#) ([snake](#) &Obj)
przesun pola tablicy segmentow o 1
- bool [goSnake](#) ([snake](#) &Obj)
1 krok weza w dobra strone
- bool [selfColl](#) ([snake](#) &Obj)
kolizja ze soba

4.7.1 Macro Definition Documentation

4.7.1.1 SCREEN_HEIGHT

```
#define SCREEN_HEIGHT 64
```

Definition at line 8 of file [snake.h](#).

4.7.1.2 SCREEN_WIDTH

```
#define SCREEN_WIDTH 128
```

Definition at line 7 of file [snake.h](#).

4.7.1.3 SnMAXsiz

```
#define SnMAXsiz 10
```

Definition at line 4 of file [snake.h](#).

4.7.2 Function Documentation

4.7.2.1 addSegm()

```
void addSegm (  
    snake & Obj,  
    int aHei,  
    int aWid)
```

dodaj segment weza

Definition at line 62 of file [snake.cpp](#).

```
00062                                     {  
00063     if(Obj.CurrLen<SnMAXsiz) {  
00064         for (int i = Obj.CurrLen; i > 0; i--) {  
00065             Obj.segs[i] = Obj.segs[i - 1];  
00066         }  
00067         Obj.segs[0].heig=aHei;  
00068         Obj.segs[0].wid=aWid;  
00069         Obj.CurrLen++;  
00070     }  
00071 }
```

4.7.2.2 changeDir()

```
void changeDir (
    snake & Obj,
    bool ifRight)
```

zmien kierunek weza

Parameters

<i>Obj</i>	ref do weza
<i>ifRight</i>	czy skret w prawo

Definition at line 11 of file [snake.cpp](#).

```
00011                                     { //true -->right
00012     if (ifRight) {
00013         //go right
00014         Obj.dir = (Obj.dir + 1) % 4; //+1
00015     } else {
00016         //go left
00017         Obj.dir = (Obj.dir + 3) % 4; //-1
00018     }
00019 }
```

4.7.2.3 goSnake()

```
bool goSnake (
    snake & Obj)
```

1 krok weza w dobra strone

Parameters

<i>Obj</i>	ref do snake
------------	--------------

Returns

if gameOver –tj. czy wyszedl poza ekran

petle musza ustawiac nowy head i kopiowac wszystkie inne elem

Definition at line 24 of file [snake.cpp](#).

```
00024                                     {
00025     moveSnake (Obj);
00026
00027     switch (Obj.dir*3) {
00028
00029     case 12: //gora
00030     case 0 : //gora
00031         Obj.segs[0].heig--;
00032         // heig--; //na OLED -- idzie w gore,
00033         break;
00034
00035     case 3 : //prawo
00036         Obj.segs[0].widt++;
00037         // widt++;
00038         break;
00039
00040     case 6 : //dol
00041         Obj.segs[0].heig++;
00042         // heig++;
```

```

00043         break;
00044
00045     case 9 : //prawo
00046         Obj.segs[0].widt--;
00047         // widt--;
00048         break;
00049     }
00050
00051     //czy snake wyszedl poza ekran
00052     if(Obj.segs[0].heig > SCREEN_HEIGHT || Obj.segs[0].heig < 0)
00053         return true;
00054     else if(Obj.segs[0].widt>SCREEN_WIDTH || Obj.segs[0].widt < 0)
00055         return true;
00056     else {
00057         return false;
00058     }
00059     // displSnake(2);
00060 }

```

4.7.2.4 moveSnake()

```

void moveSnake (
    snake & Obj)

```

przesun pola tablicy segmentow o 1

Definition at line 73 of file [snake.cpp](#).

```

00073     {
00074         if(Obj.CurrLen<SnMAXsiz) {
00075             for (int i = Obj.CurrLen-1; i > 0; i--) {
00076                 Obj.segs[i] = Obj.segs[i - 1];
00077             }
00078         }
00079     }

```

4.7.2.5 selfColl()

```

bool selfColl (
    snake & Obj)

```

kolizja ze soba

Returns

true dla kollizji – game Over;

najpierw dla każdego elementu porównujemy wysokość, zamiast wys && szer, dzięki temu możemy odciąć połowę operacji od razu – szybsze przetwarzanie

Definition at line 81 of file [snake.cpp](#).

```

00081     {
00082         for(int i=1; i<Obj.CurrLen-1; i++) {
00083             if(Obj.segs[0].heig==Obj.segs[i].heig ) {
00084                 if(Obj.segs[0].widt==Obj.segs[i].widt) {
00085                     return true;
00086                 }
00087             }
00088         }
00089         return false;
00090     }

```

4.7.2.6 setSeg()

```
void setSeg (
    snake & Obj,
    int seg,
    int aheig,
    int awidt)
```

ustaw parametry weza

Parameters

<i>Obj</i>	ref do weza
<i>seg</i>	numer segmentu weza
<i>aheig</i>	wysokosc do ustawienia
<i>awidt</i>	szer do ustawienia

Definition at line 4 of file [snake.cpp](#).

```
00004
00005     if(seg<SnMAXsiz) {
00006         Obj.segs[seg].heig=aheig;
00007         Obj.segs[seg].widt=awidt;
00008     }
00009 }
```

4.8 snake.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 //max rozmiar weza
00004 #define SnMAXsiz 10
00005
00006 //wymiary ekranu OLED
00007 #define SCREEN_WIDTH 128
00008 #define SCREEN_HEIGHT 64
00009
00013 struct snakeSeg{
00014     // uint_8
00015     int heig=0;
00016     int widt=0; //pozycja segm
00017 };
00018
00022 struct snake{
00023     snakeSeg segs[SnMAXsiz];
00024     int CurrLen=3;
00025     int dir = 0; // 0 lub 4-gora, 1-prawo, 2-dol, 3-lewo
00026 };
00027
00035 void setSeg(snake &Obj, int seg, int aheig, int awidt);
00036
00037
00043 void changeDir (snake &Obj, bool ifRight);
00044
00048 void addSegm(snake &Obj, int aHei, int aWid);
00049
00053 void moveSnake(snake &Obj);
00054
00055
00061 bool goSnake (snake &Obj);
00062
00068 bool selfColl (snake &Obj);
```